

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 5: Primary and Secondary Growth: How Do Plants Show Growth?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	C	B	C	D	A	B

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on which parts of the seedlings are getting longer

Q1. A farmer in Kakamega watches young maize and bean seedlings after germination. Which parts of each seedling are mainly getting longer?

- A. The whole stem and root grow longer evenly, like an animal growing all over
- B. Only the bark between the xylem and phloem
- C. The tips of the shoot and the tips of the root**
- D. Only the old leaves near the soil

Correct answer: C. Growth in length (primary growth) happens at the shoot and root tips, where the apical meristems are. Plants do not grow evenly all over like animals.

USE AFTER: OBSERVE PHASE — Station 1: prepared slide or photomicrograph of young dicot (bean) and monocot (maize) stems

Q2. A learner views a low-power section of a young bean stem and a young maize stem under a microscope. How are the vascular bundles arranged?

- A. Both stems have bundles in a ring
- B. Bean: scattered through the stem. Maize: in a ring
- C. Bean: in a ring. Maize: scattered through the stem**
- D. Both stems have a single bundle in the centre

Correct answer: C. The young dicot (bean) stem has its vascular bundles in a ring, while the monocot (maize) stem has scattered bundles.

USE AFTER: OBSERVE PHASE — Station 2: cut Eucalyptus/Cypress/Grevillea log showing annual rings; count rings and record

Q3. A learner counts 14 rings on a cut Eucalyptus log. What do the rings and the thick trunk show about the tree?

- A. The trunk has grown only in length through primary growth
- B. The trunk has increased in girth through secondary growth**
- C. The trunk has thickened by stretching of existing cells only
- D. The trunk has no cambium

Correct answer: B. Annual rings come from new xylem made by the vascular cambium. This is secondary growth, which increases girth.

USE AFTER: EXPLAIN PHASE — CER: Why do bean plants thicken and branch while maize stems stay mostly the same width?

Q4. Bean plants thicken and branch, but maize stems stay mostly the same width. What is the best explanation?

- A. Maize stems have a cambium ring but it is too thin to work
- B. Bean stems thicken only because they take in more water
- C. Maize stems have scattered vascular bundles with no cambium, so they do not thicken much**
- D. Maize does not have a shoot tip, so it cannot grow

Correct answer: C. Secondary growth needs a vascular cambium. The bean (dicot) stem has cambium between xylem and phloem. The maize (monocot) stem has scattered bundles and no cambium, so it mostly grows only in length.

USE AFTER: EXPLAIN PHASE — Build the 3-step model of secondary growth in a woody dicot (Step 2: cambium forms a complete ring)

Q5. In a 3-step model of secondary growth in a Grevillea stem, the cambium forms a complete ring. What does this ring of cambium produce?

- A. Only cork cells on both sides
- B. New root tips at the base of the stem
- C. Secondary phloem inward and secondary xylem outward
- D. Secondary xylem inward and secondary phloem outward**

Correct answer: D. The vascular cambium makes secondary xylem towards the inside and secondary phloem towards the outside. This adds to the girth of the stem.

USE AFTER: MODEL BUILDING PHASE — Revise Day 1 model: 'After germination: how the plants grow' with labelled arrows for apical and lateral meristems

Q6. A learner revises a model of maize (hypogeal) and bean (epigeal) seedlings after germination. Which statement should be in the model?

- A. Both seedlings start growing in length by primary growth from the shoot and root apical meristems**
- B. Maize grows only by secondary growth, and bean grows only by primary growth
- C. Neither seedling has any meristem
- D. Bean seedlings start with secondary growth, but maize seedlings start with primary growth

Correct answer: A. Both maize and bean begin growth after germination by primary growth at the shoot and root apical meristems. Only the bean (dicot) later adds secondary growth.

USE AFTER: END OF LESSON — End of lesson: teacher mini-explanation and quick quiz on primary and secondary growth

Q7. Which pair correctly matches the type of growth with the meristem and what increases?

- A. Primary growth: lateral meristems, girth increases
- B. Primary growth: apical meristems, length increases**
- C. Secondary growth: apical meristems, length increases
- D. Secondary growth: apical meristems, girth increases

Correct answer: B. Primary growth happens at apical meristems and increases length. Secondary growth happens at lateral meristems (vascular cambium and cork cambium) and increases girth.